

A MATTER OF MATTER

DESCRIPTION: Students will be asked questions or will conduct experiments at stations as they relate to the properties of matter. (Safety Goggles are required.)

NUMBER OF PARTICIPANTS: Maximum of 2

IMPOUND: NO

COMPETITORS BRING:

- Writing utensil;
- Safety goggles
- 3-ring binder of any size with any research/reference material. The binder may contain any item, such as a book, computer generated printout, or student created paperwork. No electronic materials are allowed. All materials must be securely bound inside the binder, so that when it is opened vertically (upside down) and given a light shake test no materials will fall out.

THE COMPETITION:

1. Each team will move from one station to another. There will be at least 4 stations.
2. Team will either be asked questions or will be required to complete experiments or make observations as they relate to the properties of matter.
3. The topics that may be tested or in which an experiment may be conducted include but are not limited to:
 - a. Compare and contrast the physical properties of matter (solids, liquids, and gases) such as mass, shape, volume, color, texture, temperature, attraction to magnets, and hardness
 - b. Describe the changes water undergoes when it changes state through heating and cooling (such as melting, freezing, boiling, evaporation, and condensation)
 - c. Describe the differences between mixtures, solutions, and compounds
 - d. Identify solutes, solvents, and solutions from given examples
 - e. Describe or be asked to separate a given set of mixtures, or solutions using materials provided by the event supervisor
 - f. Determine what will affect the rate of a solid dissolving in a liquid, including temperature, concentration, agitation rate, particle size, that can be tested in a lab. Pressure will not be tested.
 - g. Identify physical versus chemical changes or conduct an experiment to identify and measure the changes
4. Teams may be asked to do experiments dealing with sink/float, changes to states of matter, mixtures/solutions, dissolving rates, etc

RELATED FLORIDA STANDARDS:

- SC.5.P.8.1 Compare and contrast the basic properties of solids, liquids, and gases, such as mass, volume, color, texture, and temperature
- SC.3.P.8.1 Measure and compare temperatures of various samples of solids and liquids
- SC.3.P.8.2 Measure and compare the mass and volume of solids and liquids
- SC.3.P.8.3 Compare materials and objects according to properties such as size, shape, color, texture, and hardness
- SC.4.P.8.1 Measure and compare objects and materials based on their physical properties including mass, shape, volume, color, hardness, texture, odor, taste, attraction to magnets.

- SC.5.P.8.3 Demonstrate and explain that mixtures of solids can be separated based on observable properties of their parts such as particle size, shape, color, and magnetic attraction
- SC.5.P.8.2 Investigate and identify materials that will dissolve in water and those that will not and identify the conditions that will speed up or slow down the dissolving process
- SC.5.P.9.1 Investigate and describe that many physical and chemical changes are affected by temperature
- SC.3.P.9.1 Describe the changes water undergoes when it changes state through heating and cooling by using familiar scientific terms such as melting, freezing, boiling, evaporation and condensation
- SC.4.9.1 Identify some familiar changes in materials that result in other materials with different characteristics, such as decaying animal or plant matter, burning, rusting, and cooking

SCORING:

1. High score wins
2. Each question or station will be assigned a predetermined set of points by the event supervisor and communicated to teams
3. Ties will be broken by predetermined tie-breaker questions which are communicated to teams